Roll No. ....

Total Pages : 2

004401

May 2023

B.Tech. (CSE/IT) - IV SEMESTER

Organizational Behaviour (HSMC-03)

Time : 3 Hours

Max. Marks : 75

Instructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a queestion are to be attempted adjacent to each other.*

PART–A

1. Briefly discuss the following :

(a) Organizational climate. (1.5)

(b) Importance of organizational behaviour. (1.5)

(c) Attitude. (1.5)

(d) Organizational development. (1.5)

(e) Factors in working environment which lead to stress. (1.5)

(f) Work team. (1.5)

(g) Reinforcement.

(h) Managerial ethics. (1.5)

(i) Group cohesiveness. (1.5)

(j) Role of personality in predicting employee behaviour. (1.5)

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PART–B

2. (a) Discuss the various functions of Management. (7.5)

   (b) Explain system and quality viewpoint of management. (7.5)

3. (a) Explain the concept and process of perception.(7.5)

   (b) Critically examine Maslow's need hierarchy model of motivation. (7.5)

4. (a) State and explain the different leadership styles. (7.5)

   (b) Describe different stages of group formation. (7.5)

5. (a) "Culture of an organization influence employee behaviour". Explain. (7.5)

   (b) Why is organizational change often resisted by individuals and groups within the organization? How can such resistance be prevented or overcome? (7.5)

6. (a) What is management information system? Why it is required in the organization? (7.5)

   (b) Explain various levels of management and skills required at these levels. (7.5)

(a) Explain various organizational structures and their effect on human behaviour. (7.5)

(b) Explain various sources and types of conflicts. (7.5)

Roll No. .... Total Pages : 4

003401

May 2023

B.Tech. (CE/IT/CSE(AIML)/ 4th Semester

Discrete Mathmeatics (PCC-CS-401/PCC-CSH-401)

Time : 3 Hours] [Max. Marks : 75

Instructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART–A

1. (a) From the given sets, which set(s) follow the Well Ordering Principle–Set of Natural numbers, Set of Real numbers, Set of Even numbers, Set of integers. (1.5)

(b) Three red balls are to be placed in ten indistinguishable boxes but one box can contain exactly one ball. Find the number of distinct ways in which the balls can be placed? (1.5)

(c) Every student of a class participates in at least one of the games being played in the school but none of the students play all the three games. Make the Venn diagram for this. (1.5)

(d) Define monoid in relevance to Algebraic Structure? (1.5)

(e) What do you mean by Planar Graph? (1.5)

(f) What do you mean by Tautology? (1.5)

(g) What do you mean by Connected Directed Graph? (1.5)

(h) In a group of 50 people, How many minimum number of people have their birthday in the same month? (1.5)

(i) What do you mean by Ternary Relation? (1.5)

(j) What do you mean by one-to-one correspondence in the functions. (1.5)

PART–B

2. (a) If $f(x)$ and $g(x)$ be the function from these the set of integers defined as $f(x) = 2x^2 + 2$ and $g(x) = 3x + 3$ then determine following :

(i) fog of

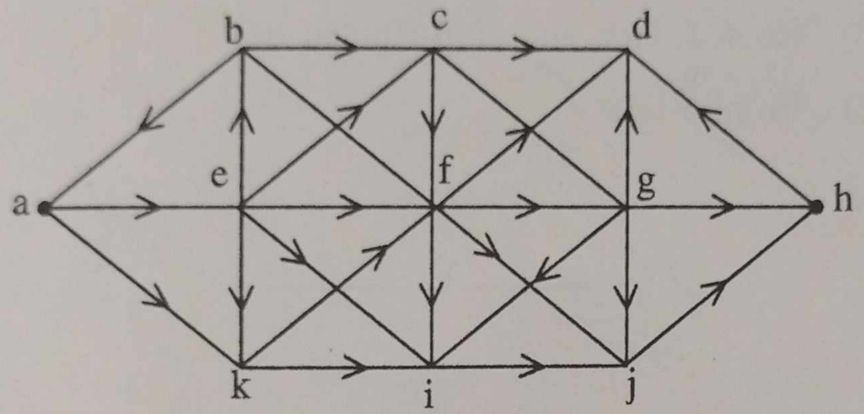
(ii) go fog

(10)

(b) If $f(x)$ be a function from the set $\{1, 2, 3, 4\}$ to the set $\{a, b, c, d]$ with $f(1) = a, f(2) = b, f(3) = c, f(4) = d$, then determine $f^{-}(a), f^{-}(b), f^{-}(c), f^{-}(d)$. (5)

3. (a) What do you mean by prefix codes? (5)

(b) Determine the chromatic number for the given graph. (10)



4. Obtain the, both, CNF and DNF for the given Boolean Function without using the truth table– $(x + y')z$. (15)

5. (a) If R is a binary relation on the set of integers i.e. {..., –2, –1, 0, 1, 2, 3, ...} such that (a, b) G R if and only if $(a - b) \geq 1)$ then determine whether the relation R is (10)

   (i) Reflexive.

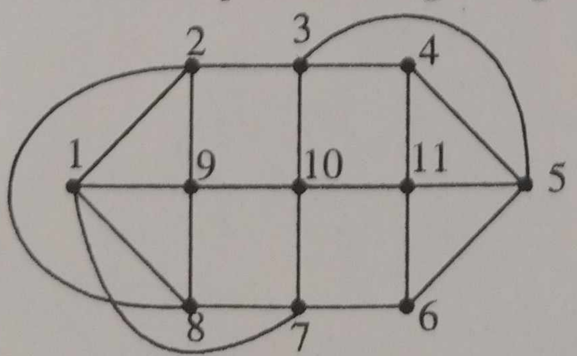
   (ii) Symmetric.

   (iii) Antisymmetric.

   (iv) Transitive.

   (b) Define Partial Ordering Relation and Equivalence Relation. (5)

6. (a) Find the Eulerian path in the given graph. (10)



(7)

7. Write short note on the following : (15)

(a) Hamiltonian Circuit.

(b) Rooted Tree and Non-Rooted tree.

(c) Infinite set.

Roll No. .... Total Pages : 3

003402

May, 2023

B.Tech. (CE/IT/AIML/CSE(AIML) IV SEMESTER

Computer Organization & Architecture

(PCC-CS-402/PCC-CSH-402)

Time : 3 Hours] [Max. Marks : 75

Instructions:

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART-A

1. (a) Write any three addressing modes with examples. (1.5)

(b) What is the use of Instruction Register? (1.5)

(c) What do you mean by Program Status Word? (1.5)

(d) Write down the binary value of $(-7)_{10}$, $(-8)_{10}$, $(+3)_{10}$ if 2's complement representation is used? (1.5)

(e) Explain write-through and write-back policy of cache memory. (1.5)

(f) Explain the I/O interface named as USB. (1.5)

(g) Subtract $(FF)_{16}$ from $(AB)_{16}$. (1.5)

(h) What do you mean by privileged and non-privileged instructions? (1.5)

(i) What do you mean by Cache Coherency? (1.5)

(j) Write any three assembly level instructions for a particular processor and explain their use. (1.5)

PART-B

2. (a) Explain the block diagram of general-purpose CPU. (10)

   (b) How an instruction is interpreted using RTL. (5)

3. (a) Differentiate between hardwired and micro-programmed control unit. (5)

   (b) Explain Booth Multiplier algorithm. (10)

4. Explain the architecture of 8086 microprocessor. (15)

5. (a) What do you mean parallel processing? (5)

   (b) What do you mean by pipelining? Explain with diagram. Write down the formula to calculate the throughput and speedup. (10)

6. (a) How to multiply two numbers using shift and add instructions? (10)

(b) Differentiate between Cache Size and Block Size. (5)

7. Write short notes on the following : (15)

(i) Ripple Carry Adder.

(ii) Software Interrupts.

(iii) Associative memory.

Roll No. .... Total Pages : 4

003403

May, 2023

B.Tech. [CE/IT/CSE(AIML))/(CE(Hindi Medium)]

IV SEMESTER

Operating System (PCC-CS-403/PCC-CSH-403)

Time : 3 Hours] [Max. Marks : 75

Instructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART–A

1. (a) Define multiprogramming. (1.5)

(b) What is the difference between long-term scheduler and short-term scheduler? (1.5)

(c) What is the difference between non-preemptive and preemptive scheduling? (1.5)

(d) What is a counting semaphore? Explain with *one* example. (1.5)

(e) What is the difference between internal and external fragmentation? (1.5)

(f) What is demand paging? (1.5)

(g) How do you prevent circular wait condition in deadlock? (1.5)

(h) What is Belady's anomaly? Explain with *one* example. (1.5)

(i) What is double buffering? (1.5)

(j) What is multithreading? (1.5)

PART–B

2. (a) Explain the architecture of a microkernel based operating system. (10)

   (b) Explain 7-state process state diagram including suspended processes. (5)

3. (a) Write the algorithms for providing synchronization solution to dining-philosopher problem using semaphore. The solution must be free from deadlok/livelock. (5)

   (b) Explain the Banker's algorithm to avoid deadlock. (10)

4. Consider the following scenario of processes with their priority. (5)

Process	Arrival time	Execution time	Priority
P1	0	12	5
P2	2	25	1
P3	3	3	3
P4	5	9	4
P5	6	13	2

Draw the Gantt chart for the execution of the processes showing their start time and end time using FCFS, priority-number based scheduling (preemptive), SRN (without considering the priority) RR with time quantum = 8. Calculate turnaround time, normalized turnaround time, waiting time for each process and average turnaround time, average normalized turnaround time and average waiting time for the system.

5. (a) Explain the hierarchical page table structure. (5)

(b) Consider a system with following information. (10)

Process	Alloc			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
PI	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

(i) What is the content of matrix need?

(ii) Is the system in safe state?

(iii) A request from process P1 arrives for [1 0 2], can the request be granted immediately?

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6. (a) Consider the following page reference string :

1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. Implement FIFO, Optimal, LRU page replacement algorithms and compare the performance based on the number of page faults with frame size 3 and 4. (10)

(b) Discuss the Indexed file allocation method in detail. (5)

7. (a) Explain SCAN and C-SCAN disk scheduling algorithms with example and diagrams. (10)

(b) Explain the layered structure of I/O software. (5)

Roll No. .... Total Pages : 5

003404

May 2023

B.Tech. - IV SEMESTER

Design & Analysis of Algorithms (PCC-CS-404)

Time : 3 Hours] [Max. Marks : 75

Instructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART–A

1. (a) What is the time complexity of the following code? Justify your answer.

```
int i, j, k = 0;
  for (i = n / 2; i <= n; i++) {
    for (j = 2; j <= n; j = j * 2) {
      k = k + n/2;
  }}
```

(1.5)

(b) Sort the following functions in the decreasing order of their asymptotic (big-0) complexity : $f_1(n) = n^{\sqrt{n}}$, $f_2(n) = 2^n$, $f_3(n) = (1.000001)^n$, $f_4(n) = n^{10} * 2^{(n/2)}$. (1.5)

211 [P.T.O.

(c) Differentiate Greedy algorithm and Dynamic programming. (1.5)

(d) State Job Sequencing with Deadline Problem. Write down time and space complexity if problem solved by Greedy approach. (1.5)

(e) Define Principle of Optimality with suitable example. (1.5)

(f) Draw state space tree of 4-Queens problem. (1.5)

(g) Why does Dijkstra's algorithm fail on negative weights? (1.5)

(h) Draw binary search trees for the given set of keys and their corresponding frequencies and find the Optimal among them. keys[ ] = {10, 12, 20}, freq[ ] = {34, 8, 50} (1.5)

(i) Explain Least Cost Search function for branch and bound algorithm design technique. (1.5)

(j) What is the importance of approximation algorithm? (1.5)

PART–B

2. (a) Solve the following recurrence relation.

(i) $T(n) = 2T(n/2) + \log n$ with $T(1)=1$

(ii) $T(n) = 2T(\sqrt{n}) + 1$ if $n > 2$ and $T(n) = 2$ if $0<n<=2$. (10)

(b) How are time and space trade-offs used to optimize the performance of an algorithm? Provide an example of an algorithm that optimizes time at the expense of space and vice-versa. (5)

3. (

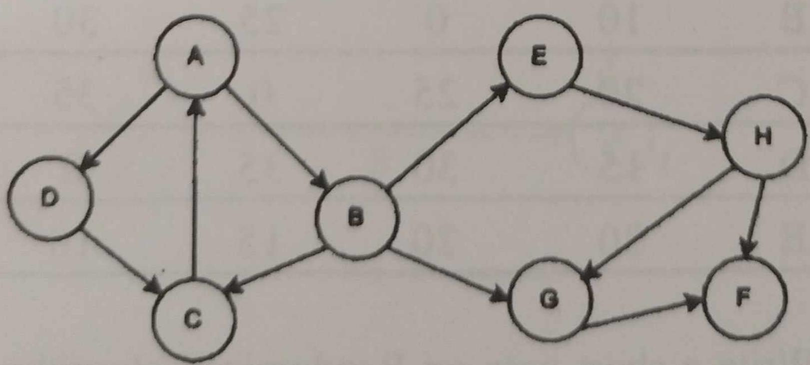
3. (a) Give a dynamic-programming solution to the 0-1 knapsack problem that runs in 0(n/W) time, where n is the number of items and W is the maximum weight of items that the thief can put in his knapsack.

Consider the weights and values of items listed below. The task is to pick a subset of these items such that their total weight is no more than 5 Kgs and o their total value is maximized.

Item No.	Weight (Kg)	Values (Rs.)
1	2	3
2	3	4
3	4	5
4	5	6

(b) Consider the given graph. In what order will the nodes be visited using a Breadth First Search and Depth First Search? (Assume starting vertex A) (5)

Note: If a node has multiple neighbors then select which is alphabetically near to node.



4. (a) The N-Queen problem is a classic problem in computer science, where the goal is to place N queens on an N × N chessboard so that no two queens attack each other.

(i) Write a brute-force algorithm to solve the N-Queen problem. Analyze the time complexity of your algorithm, and explain why it is not efficient for large values of N.

(ii) Write a backtracking algorithm to solve the N-Queen problem. Analyze the time complexity of your algorithm, and compare it with the brute-force algorithm. (12)

(b) Describe the Traveling Salesman Problem and explain why it is NP-complete. (3)

5. (a) A delivery truck travels between multiple destinations. The truck starts at city A and visits cities B, C, D, and E before returning to A. Design an algorithm to find the shortest path for the truck to travel while minimizing cost. Use following distance matrix to solve the problem. (8)

	A	B	C	D	E
A	0	10	20	15	30
B	10	0	25	30	20
C	20	25	0	35	15
D	15	30	35	0	10
E	30	20	15	10	0

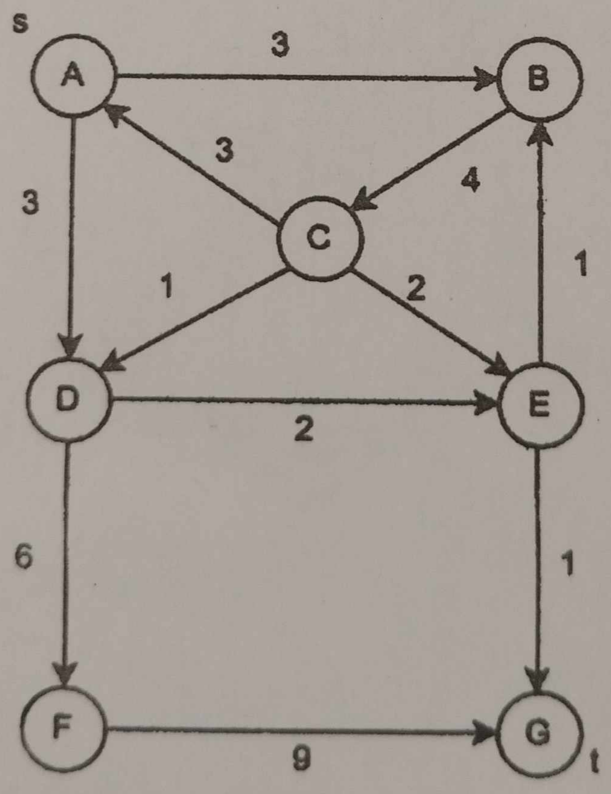
(b) Write a short note on Randomized algorithms. (7)

6. (a) Define spanning tree. Write Kruskal's algorithm for finding minimum cost spanning tree. Describe how Kruskal's algorithm is different from Prim's algorithm for finding minimum cost spanning tree. (10)

(b) What is Topological Sorting? Explain with example. (5)

7. (a) What is the relationship among P, NP, NP-Hard and NP-Complete problems? Show with the help of a diagram. (8)

(b) In a given directed graph with source s and sink t, where each edge in the graph has a non-negative capacity. Find the maximum flow that can be sent from s to t. (7)



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May 2023

B.Tech - IV SEMESTER

Signal & Systems - (ECC01)

Time: 3 Hours

Max. Marks:75

Instructions:

1. *It is compulsory to answer all the questions (1.5 marks each) of Part -A in short.*
2. *Answer any four questions from Part -B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART –A

Q1 (a) Plot the following signal : $2u(t+2) - 2u(t-3)$. (1.5)

(b) Find whether the signal $\sin\left(\frac{\pi}{3}n\right)$ is a power signal or energy signal. Also find the energy and power of the signal. (1.5)

(c) If Fourier transform of $x(t)$ is $X(j\omega)$, then show that Fourier transform of $x(t+T) + x(t-T)$ is equal to $2X(j\omega)\cos\omega t$. (1.5)

(d) Write the conditions for convergence of Fourier series. (1.5)

(e) State how ROC of Laplace transform is useful in determining stability of systems. (1.5)

(f) Compare Fourier, Laplace transform and z – transform. (1.5)

(g) State Parseval's theorem. (1.5)

(h) Find the Nyquist rate & Nyquist interval for the following signal (1.5)

$x(t) = -10\sin(40\pi t)\cos(300\pi t)$

(i) Determine the system function of a discrete – time system described by the difference equation $y(n) - \frac{1}{3}y(n-1) + \frac{1}{5}y(n-2) = x(n) - 2x(n-1)$. (1.5)

(j) When does aliasing occur ? How can it be avoided? (1.5)

PART –B

Q2 (a) Give an account for the classifications of signals in detail. State whether the system $y(t) = t\,x(-t)$ is causal, stable, linear and time invariant. (10)

(b) Find the convolution sum between $x(n) = \{1,4,3,2\}$ and $h(n) = \{1,3,2,1\}$ by graphical method. (5)

Q3 (a) A system is described by the differential equation (7.5)

$$\frac{d^2y(t)}{dt^2} + 6\frac{dy(t)}{dt} + 8\,y(t) = \frac{dx(t)}{dt} + x(t)$$

with initial conditions $\frac{dy(0)}{dt} = 3$, $y(0) = 1$ and input $x(t) = u(t)$. Find the transfer function and the output signal $y(t)$

(b) Determine the poles and zeros of the rational function of s given as

$X(s) = \frac{(s+1)(s^2+10s+41)}{(s+4)(s^2+4s+13)}$ and also sketch pole-zero plot. (7.5)

Find the Laplace transform of $\sin^2 3t\ u(t)$.

Q4 (a) Using suitable z-transform properties, find X(z) of

$$x(n) = (n-2)\left(\frac{1}{2}\right)^{(n-2)} u(n-2)$$ (7.5)

(b) Solve the following difference equation using z-transform

$$y(n) + 2y(n-1) = x(n)$$ (7.5)

with $x(n) = \left(\frac{1}{3}\right)^n u(n)$ and the initial condition $y(-1) = 1$.

Q5 (a) Compute the circular convolution of the following two sequences using DFT $x_1(n) = \{0,1,0,1\}$ and $x_2(n) = \{1,2,1,2\}$. (7.5)

(b) The system produces the output of $y(t) = e^{-t}\ u(t)$ for an input of $x(t) = e^{-2t}\ u(t)$. (7.5) Determine the impulse response and frequency response of the system.

Q6 (a) Discuss the characterization of causality and stability of Linear shift–invariant (7.5) system. Perform convolution of the following causal signals:

$$x1(t) = \cos t\ u(t); \quad x2(t) = t\ u(t)$$

(b) State and prove time reversal property of DTFT. Find the DTFT of the following (7.5) discrete time signal : $\cos\left(\frac{\pi n}{2}\right) u(n)$

Q7 (a) *What are the advantages of state space analysis? Find the state transition matrix for* (7.5) *the continuous time system parameter matrix,* $A = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix}$

(b) State and explain sampling theorem both in time and frequency domain with (7.5) necessary quantitative analysis and illustrations.

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Roll No. .... Total Pages : 3

003406

May 2023

B.Tech. (CE/CE(DS)/CSE(AIML))/CE
(Hindi Medium)) 4th Semester

ENVIRONMENTAL SCIENCES (MC-03/MCH-03)

Time : 3 Hours] [Max. Marks : 75

Instructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART-A

1. Write a short note on the following :
   - (a) Green marketing. ($1^1/_2$)
   - (b) Eutrophication. ($1^1/_2$)
   - (c) Ocean Thermal Energy Conservation (OTEC). ($1^1/_2$)
   - (d) Ecological succession. ($1^1/_2$)
   - (e) Biogeographic Zones of India. ($1^1/_2$)
   - (f) Hotspot of biodiversity. ($1^1/_2$)
   - (g) Endemic species. ($1^1/_2$)

(h) Sanitary landfills. $(1^1/_2)$

(i) Sustainable Development. $(1^1/_2)$

(j) Watershed Management. $(1^1/_2)$

PART-B

2. (a) How would environmental awareness help to protect our environment, explain briefly? (5)

   (b) Write an explanatory note on the multidisciplinary nature of environmental science. Briefly describe, how computer engineering plays an important role in conservation and protection of environmental quality? (10)

3. (a) Give a detailed account of various impacts of traditional and modern agricultural practices on environment. $(7^1/_2)$

   (b) What are the environmental impacts associated with mineral extraction? Briefly explain the remedial measures to reduce the impacts of mining. $(7^1/_2)$

4. (a) Define ecosystem. Give a detailed account of structural and functional attributes of an ecosystem. (10)

   (b) Why the concept of food web is more realistic than the concept of food chain? (5)

5. (a) What do you understand by the term biodiversity? Discuss the concept of biodiversity at different hierarchical levels. $(7^1/_2)$

(b) Elucidate different approaches of biodiversity conservation along with their merits and demerits. $(7^1/_2)$

(a) What is water pollution? Briefly explain the effects and various control measures of water pollution. $(7^1/_2)$

(b) What are wastelands. Give a detailed account of various methods or techniques for reclamation of wastelands. $(7^1/_2)$

Write short note on any *three* of the following :

(a) Environmental ethics.

(b) Life expectancy.

(c) Role of IT in environmental and health.

(d) Universal Declaration of Human Rights. (5×3=15)

Roll No. .... Total Pages : 3

002407

May, 2023

B.Tech. IV SEMESTER

Biology (BSC-01)

Time: 3 Hours] [Max. Marks. : 75

Instructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART-A

1. (a) Explain carbohydrates. (1.5)

(b) 5 differences between mitosis and meiosis. (1.5)

(c) Explain a note on eukaryotes. (1.5)

(d) Explain Phospholipids. (1.5)

(e) 5 differences between DNA and RNA. (1.5)

(f) Compare camera and eye with figure. (1.5)

(g) Explain Mendels Law. (1.5)

(h) Explain genetic code with figure of triplet codons.(1.5)

(i) What is oxidative and substrate level phosphorylation. (1.5)

(j) Why should an engineer study biology? Explain citing examples. (1.5)

PART-B

2. (a) Explain Glycolysis in detail, its site of action, enzymes involved, and end product. (10)

(b) What are model organism? Explain with examples and its usage in biology. (5)

3. (a) What are the different methods of isolating and identifying microbes. (5)

(b) What are biomolecules? Explain in detail diverse proteins, its types, and structures studied. (10)

4. Distinguish between, cyclic and non cyclic phosphorylation, light and dark reactions, catabolism and anabolism. (15)

5. (a) Explain in detail blood group. What are autosomal and recessive disorders? (5)

(b) Explain enzymes, its classification and mechanism of action hypothesis. (10)

6. (a) In detail explain TCA cycle with enzymes, end product and well labelled figure. (10)

(b) Explain in details basis techniques used during the process of sterilization. (5)

7. Explain in Detail photosynthesis with equation, site and well labelled figures. (15)
